

Generative AI

L-T-P-C: 0-0-2-2

1. Course Introduction

This course students to the foundational principles of Generative AI, offering a comprehensive understanding of both classical and modern generative models. It covers the theoretical concepts underpinning generative modeling and delves into practical applications, enabling students to grasp how these models work and where they can be applied. Special attention is given to open-source frameworks such as TensorFlow, ensuring that students gain hands-on experience with tools widely used in the industry. Additionally, this part introduces students to Natural Language Processing (NLP) fundamentals, bridging the gap between generative models and real-world textual data analysis. The integration of theoretical learning with practical tasks ensures that students develop a solid foundation in generative AI.

2. Course Outcomes

By the end of this course, participants will be able to:

1. Understand the fundamentals of generative modeling and its applications.
 2. Implement and evaluate basic generative models like VAEs and GANs.
 3. Utilize open-source tools for generative model development.
 4. Apply basic NLP techniques to preprocess and analyze textual data.
 5. Recognize ethical considerations in generative AI applications.
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3. COURSE PREREQUISITES:

- Basic knowledge of Python programming.
- Understanding of machine learning fundamentals, including supervised and unsupervised learning.
- Familiarity with linear algebra, probability, and calculus.

4. Course Content:

UNIT-I	Introduction to Generative AI	09 Hours
Overview of Generative AI, Generative vs. Discriminative Models, Probability Fundamentals (Likelihood, Prior, Posterior), Introduction to Classical and Deep Generative Models, Open-Source Tools (PyTorch, TensorFlow), Applications of Generative AI.		
UNIT-II	Generative Techniques & Autoregressive Models	06 Hours
Autoregressive Models (e.g., PixelRNN, PixelCNN), Sampling Techniques (MCMC, Gibbs Sampling), Challenges in High-Dimensional Data, Open-Source Autoregressive Implementations.		
UNIT-III	GANs and Diffusion Models	10 Hours
GAN architecture and minimax objective. Training dynamics and mode collapse. Key architectures: DCGAN, StyleGAN. Conditional GANs. WGAN and improved objectives. GAN evaluation metrics. Domain translation: CycleGAN, Pix2Pix. Diffusion models: forward/reverse processes. DDPM fundamentals. Score-based models. Stable Diffusion and latent diffusion. Comparative analysis of generative approaches.		
UNIT-IV	Introduction to Natural Language Processing	12 Hours
Text Preprocessing, Tokenization, Stopword Removal, Introduction to Word Embeddings (Word2Vec, GloVe), Applications in Text Analysis and Language Modeling.		
UNIT-V	Advanced NLP and Transformer-Based Models	08 Hours

Advanced NLP Techniques: Transformer Architecture, BERT and Fine-Tuning, Open-Source NLP Models, Applications in Text Classification, Summarization, and Question Answering.

REFERENCE

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| 1 | https://developers.google.com/machine-learning/gan |
| 2 | https://developers.google.com/machine-learning |

HARDWARE REQUIREMENT

- Desktop/Laptop with modern multi-core CPU
- CUDA-compatible GPU with at least 8GB VRAM (16GB+ recommended for larger models)
- Minimum 16GB system RAM (32GB recommended)
- 100 GB+ fast storage (SSD recommended)
- Alternative: Access to cloud computing platforms with GPU instances

TextBooks:

- "Speech and Language Processing" by Daniel Jurafsky and James H. Martin
- "Foundations of Statistical Natural Language Processing" by Christopher D. Manning
- "Generative Adversarial Networks (GANs) in Action: Deep learning with Generative Adversarial Networks" by Jakub Langr and Vladimir Bok.
- "Generative AI for Everyone: Deep learning, NLP, and LLMs for creative and practical applications" by Karthikeyan Sabesan